



**Department of Computer Science and Engineering
University of Dhaka**

**CSE 4113: Internet Programming Lab
Assignment 1: Software Requirements Specification (SRS)**

**Project Title:
CSE SmartPortal: A Unified Web, CMS and
Academic Resource Management System**

Team Name: LogicLoop

Submitted To:

Dr. Mamun Or Rashid, Professor, CSEDU
Md. Fahim Arefin, Lecturer, CSEDU
Md. Ahasanul Alam, Lecturer, CSEDU

Submitted By:

Abdullah Al Arman Emon (50)
Shah Jamal Islam (81)
Suhail Tanvir Nahin (82)
Md. Rakib Hossain (88)
Imran Shorif Shuvo

March 7, 2026

Contents

1	Introduction	1
1.1	Purpose	1
1.2	Scope	1
1.3	Definitions, Acronyms and Abbreviations	2
1.4	References	2
1.5	Overview	2
2	Overall Description	3
2.1	Product Perspective	3
2.2	Product Functions	3
2.3	User Characteristics	3
2.4	ER Diagram	4
2.5	Constraints	5
2.6	Assumptions and Dependencies	5
3	Specific Requirements	6
3.1	Use Case Model	6
3.2	Use Case List	6
3.3	Use Case Diagram	7
3.4	System Architecture Diagram	8
3.5	Functional Requirements	8
3.5.1	CMS Requirements	8
3.5.2	Multilingual Support	8
3.5.3	Gallery Management	8
3.5.4	Scholarship Management	8
3.5.5	Event Management	8
3.5.6	Venue Reservation	9
3.6	Non-Functional Requirements	9
3.6.1	Performance	9
3.6.2	Security	9
3.6.3	Usability	9
3.6.4	SEO Optimization	9
3.6.5	Reliability	9
3.6.6	Maintainability	9
4	Supporting Information	10
4.1	Summary	10
4.2	Appendices	10

Chapter 1

Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to describe the functional and non-functional requirements of **CSE SmartPortal**, a web-based platform designed to manage the Department of Computer Science and Engineering's online presence and administrative operations.

This document provides a clear description of system features, external interfaces, user interactions, constraints, and performance requirements. It serves as a guideline for developers, testers, and stakeholders during the development of the system.

1.2 Scope

CSE SmartPortal is a unified web-based system that integrates multiple departmental services into a single platform. The system will support:

- A public-facing departmental website
- Content Management System (CMS) for non-technical staff
- News and blog publishing
- Multilingual content (Bangla/English)
- Gallery management
- Scholarship management
- Event publishing and registration
- Auditorium/Lab/Room booking system
- QR-based event check-in
- Contact and inquiry management

The system aims to reduce fragmentation by consolidating website management, scholarship administration, event coordination, and resource booking into one centralized platform.

1.3 Definitions, Acronyms and Abbreviations

Term	Description
CMS	Content Management System
SEO	Search Engine Optimization
QR	Quick Response code used for event check-in
OAuth	Authentication protocol
Admin	System administrator responsible for managing the platform
User	Any person interacting with the system (student, teacher, or visitor)

Table 1.1: Definitions, Acronyms and Abbreviations

1.4 References

1. IEEE Recommended Practice for Software Requirements Specifications (IEEE 830-1998)
2. CSEDU Project Tracks Guide
3. CSEDU Development Repository Policy
4. CSE 4113 Internet Programming Lab Course Materials

1.5 Overview

This document is organized into the following sections:

Chapter 1 – Introduction:

Provides the purpose and scope of the system.

Chapter 2 – Overall Description:

Describes the general features and constraints of the system.

Chapter 3 – Specific Requirements:

Defines detailed functional and non-functional requirements.

Chapter 4 – Supporting Information:

Includes appendices and additional supporting materials.

Chapter 2

Overall Description

2.1 Product Perspective

CSE SmartPortal will function as the **centralized digital platform** for the CSE department. The system will replace fragmented tools used for website management, event announcements, scholarship postings, and venue booking.

The system will consist of the following integrated components:

- Public Website Interface
- Admin CMS Panel
- Scholarship Management Module
- Event Management Module
- Room/Lab Reservation Module

2.2 Product Functions

The main functions of the system include:

1. Display departmental information through a **dynamic public website**.
2. Provide a **CMS dashboard** for managing website content.
3. Publish **news, announcements, and blog posts**.
4. Support **Bangla and English** language content.
5. Manage **image and video galleries**.
6. Manage **scholarship announcements and applications**.
7. Allow users to **register for events** online.
8. Provide **QR code check-in** for events.
9. Allow **booking of auditorium, classrooms, or labs**.
10. Provide **calendar view** for events and bookings.
11. Manage **contact inquiries** from website visitors.

2.3 User Characteristics

The system caters to four primary user roles, as defined in the table below:

User Type	Description
Visitor	General public viewing the website and announcements.
Student	Can apply for scholarships, register for events, and request venue booking.
Teacher	Can manage events, scholarships, and approve bookings.
Admin	Full system control including CMS, users, and content management.

Table 2.1: User Roles and Descriptions

Note: All users are assumed to have basic web browsing knowledge.

2.4 ER Diagram



Figure 2.1: Entity-Relationship (ER) Diagram for CSE SmartPortal Database

2.5 Constraints

The development and operation of the system are governed by the following constraints:

- The system must be developed as a **web-based platform**.
- The project must be maintained in a **public Git repository**.
- Authentication part
- Domain-specific email requirements:
 - Student emails: `@cs.du.ac.bd`
 - Teacher/Officer emails: `@cse.du.ac.bd`
- Support for **bilingual content** (Bangla and English).
- Accessibility via all modern web browsers.

2.6 Assumptions and Dependencies

- Users have consistent internet access and modern web browsers.
- Google authentication services (OAuth) will remain available and functional.
- The system will be hosted on a reliable web server provided by the department.
- The department will provide necessary administrative data (scholarships, events, etc.) in a timely manner.

Chapter 3

Specific Requirements

3.1 Use Case Model

Actors:

- Visitor
- Student
- Teacher
- Admin

3.2 Use Case List

Use Case	Actor	Description
View Website	Visitor	Browse website pages, news, and announcements
Manage Website Content	Admin	Create, edit, or delete pages through CMS
Publish News/Blog	Admin	Post departmental news and blog content
Manage Gallery	Admin	Upload and manage images/videos
Submit Inquiry	Visitor	Send questions through contact form
Apply for Scholarship	Student	Submit scholarship application
Review Scholarship Applications	Teacher/Admin	Review applications and publish results
Register for Event	Student/Visitor	Register for departmental events
Event Check-in	Admin	Scan QR code to confirm event attendance
Book Room/Lab	Student/Teacher	Request venue booking
Approve Booking	Admin	Approve or reject booking requests

Table 3.1: System Use Cases and Actor Interactions

3.3 Use Case Diagram

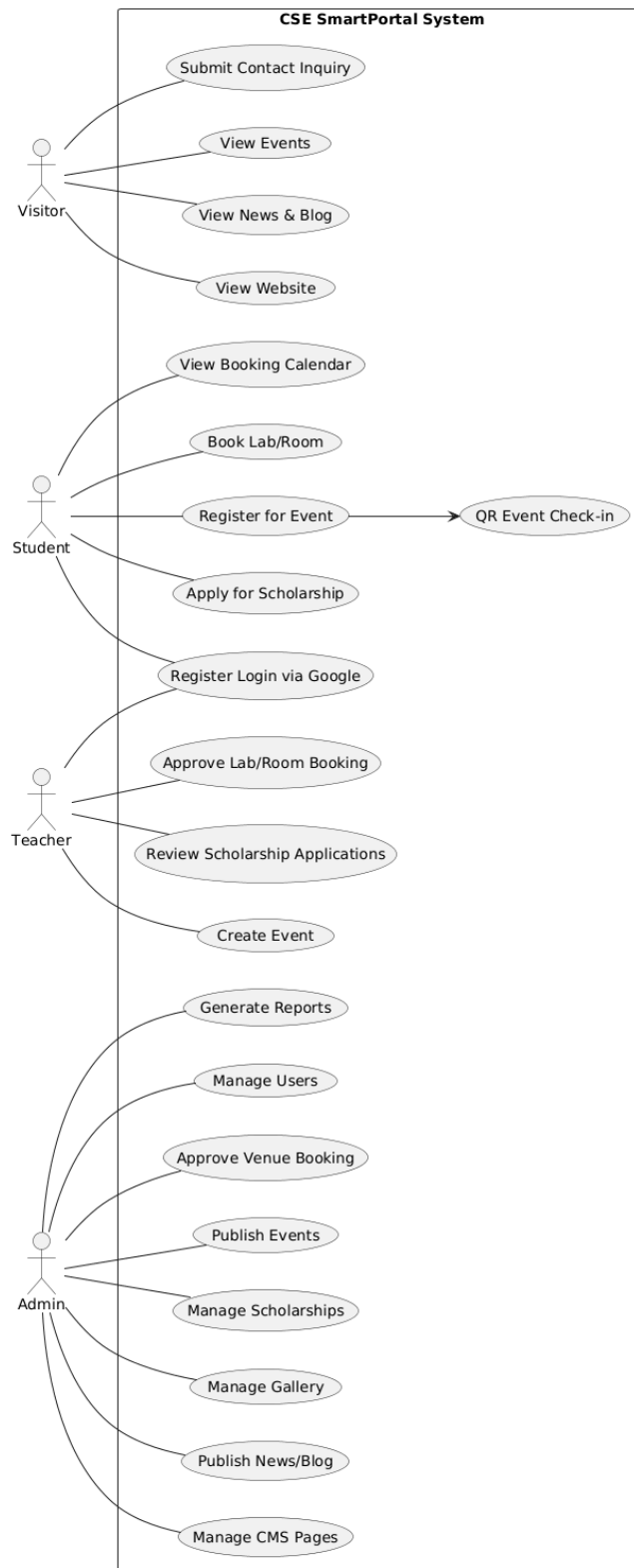


Figure 3.1: Overall System Use Case Diagram for CSE SmartPortal

3.4 System Architecture Diagram

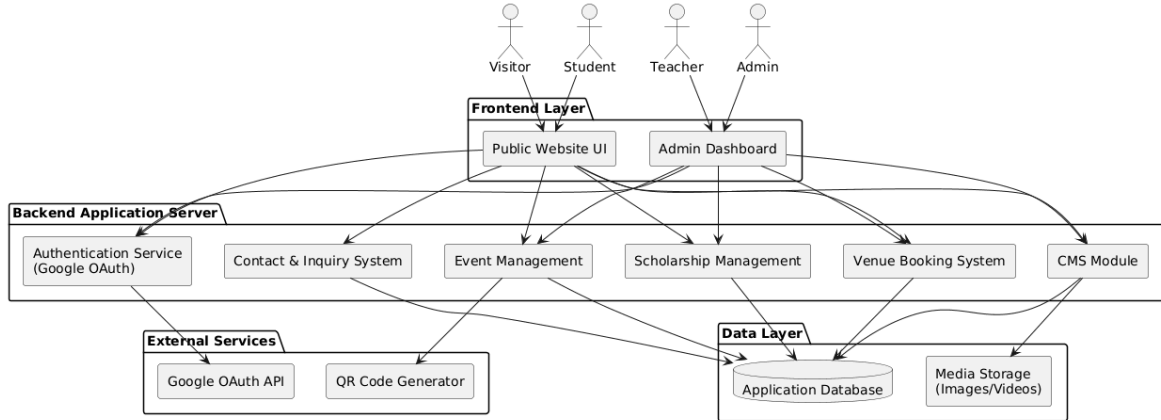


Figure 3.2: System Architecture Diagram for CSE SmartPortal

3.5 Functional Requirements

3.5.1 CMS Requirements

- The system shall allow administrators to **create, edit, and delete** website pages.
- The system shall provide a **WYSIWYG editor** for content editing.
- The system shall allow administrators to **publish blog posts and news updates**.

3.5.2 Multilingual Support

- The system shall support **Bangla and English** language content.
- The system shall allow administrators to manage **translations for pages**.

3.5.3 Gallery Management

- The system shall allow administrators to **upload images and videos**.
- The system shall **display gallery content** on the public website.

3.5.4 Scholarship Management

- The system shall allow administrators to **publish scholarship notices**.
- The system shall allow students to **submit scholarship applications**.
- The system shall allow administrators to **open or close application periods**.
- The system shall allow administrators to **review and export applications**.
- The system shall allow publishing **scholarship recipient lists**.

3.5.5 Event Management

- The system shall allow administrators to **publish events and announcements**.
- The system shall allow users to **register for events** online.

- The system shall generate **QR codes** for event check-in.
- The system shall collect **post-event feedback** from participants.

3.5.6 Venue Reservation

- The system shall allow users to **request room, lab, or auditorium booking**.
- The system shall display **booking calendar availability**.
- The system shall allow administrators to **approve or reject booking requests**.

3.6 Non-Functional Requirements

3.6.1 Performance

- The system should handle **multiple simultaneous users** without performance degradation.
- Page load times should remain under **3 seconds** under normal load conditions.

3.6.2 Security

- The system must implement **Google OAuth authentication**.
- Access permissions must be **role-based** (Visitor, Student, Teacher/Officer, Admin).
- All data transmission must occur over **HTTPS**.
- Sensitive user data must be **encrypted** in storage.

3.6.3 Usability

- The user interface should be **simple and easy to navigate**.
- The system should be **responsive** and accessible on desktop and mobile devices.
- Error messages should be **descriptive and user-friendly**.

3.6.4 SEO Optimization

- The system should support **SEO metadata** and sitemap generation.
- Pages should include **social sharing metadata** (Open Graph, Twitter Cards).

3.6.5 Reliability

- The system should maintain **consistent data storage and backups**.
- The system should target an uptime of **99%** during academic semesters.

3.6.6 Maintainability

- The codebase must be maintained in a **public Git repository** with proper documentation.
- The system should follow **modular design principles** to allow easy feature extension.

Chapter 4

Supporting Information

4.1 Summary

This Software Requirements Specification (SRS) document defines the complete set of requirements for the **CSE SmartPortal** system. The document covers:

- The purpose and scope of the system (Chapter 1)
- General product description, user types, constraints, and assumptions (Chapter 2)
- Detailed functional and non-functional requirements with use cases (Chapter 3)
- Supporting appendix materials referenced below (Chapter 4)

4.2 Appendices

Possible appendices includes:

- Use-case diagrams
- Database schema
- System architecture diagram

These materials support the understanding of the system but are not mandatory parts of the core requirements.